

TEPL 1120 – Plumbing Apprentice IB (3 Credits)

Course Description

The Plumbing IB course introduces the fundamentals of plumbing theory for the apprentice plumbers and will cover the International Plumbing Code, related math, and craft skills.

Course Objectives

- Identify fixtures, faucets and fixture fittings, water heaters, traps, interceptors and separators.
- Develop basic skills needed to read drawings and produce piping sketches.
- Apply mathematics related to plumbing and angles.
- Implement the process of making watertight joints using heat and various filler metals.
- Define machine and hand excavating with emphasis on safety.
- Explain various types of pipe and fittings used in residential and light commercial plumbing systems.
- Fabricate several piping projects in a lab situation

Course Outline

- Year IB

Textbook & Reading Materials

Mathematics for Plumbers and Pipefitters (ISBN: 9781111642600)

Modern Plumbing eBook Bundle w/2yr access code (ISBN: 9781637768303)

International Plumbing Code 2021 (ISBN: 9781609839628)

International Plumbing Code 2021 Study Companion (ISBN: 9781955052283)

Assignments & Assessments

How do I submit an assignment online?

How do I take a quiz?

How do I view my grades?

How can I use Canvas on my mobile device?

Hybrid Assignments

Work Information

Read Chapter 11: Soldering, Brazing, and Welding in the Modern Plumbing textbook

Chapter 11 Know and Understand

Chapter 11 Lab Workbook Assignment

Discussion Board: Modern Plumbing Chapter 11- Soldering, Brazing and Welding

Read Chapter 12: Excavating

Chapter 12 Know and Understand

Chapter 12 Lab Workbook Assignment

DISCUSSION: Class Roll

Study Resources

How to Braze Copper Pipe

chtt_soldering_brazing_welding.pptx

ch12_presentation_excavating.pptx

Hybrid Assignments

Read MOPL Chapter 13: Water Supply Systems

Chapter 13 Know and Understand

Chapter 13 Lab Workbook Assignment

Read MOPL Chapter 14: Water Treatment

Chapter 14 Know and Understand

Chapter 14 Lab Workbook Assignment

Discussion Board: Modern Plumbing Chapter 14 Water treatment

Study Resources

ch13_presentation_water design_systems.pptx

Water Cycle

Private Well

Ch14_presentation_water_treatment.pptx

Choosing a Water Filter Video

Water Softener Systems Video

Hybrid Assignments

Quiz: MP Chapters 13 & 14

DISCUSSION: Excavating

Read Chapter 15 Plumbing Fixtures in Modern Plumbing textbook

Chapter 15 Know and Understand

View as Student

Collapse All View Progress III

Chapter 15 Lab Workbook

Chapter 15 Fixture Activity

Discussion Board: Modern Plumbing Chapter 15 (Plumbing Fixtures)

Study Resources

ch15_plumbing_fixtures.pptx

Hybrid Assignments

Read IPC Chapter 4: FIXTURES, FAUCETS, AND FIXTURE FITTINGS

IPC 55 4 - Fixtures, Faucets and Fixture Fittings I

IPC 55 5 Quiz - fixtures, Faucets and Fixture Fittings II

DISCUSSION: Toilets

Fixture Report

Read MP Chapter 16: Piping Materials and Fittings

Chapter 16 Know and Understand Assignment

Chapter 16 Lab Workbook Assignment

Study Resources

Plumbing Fittings.pdf

ch16_Piping Materials and Fittings.pptx

Hybrid Assignments

Read Chapter 6: IPC Code Book: water supply and distribution IPC Code book

IPC SS 7 Quiz-Water Supply and Distribution I (sections 601 - 605)

IPC SS 8 Quiz - Water Supply and Distribution II (sections 606 - 613)

DISCUSSION: Class Roll

Read Chapter 17: Valves and Meters in the Modern Plumbing textbook

Chapter 17 Know and Understand Assignment

Chapter 17 Lab Workbook Assignment

Study Resources

Water supply systems.pptx

ch17 Valves and Meters.pptx

Hybrid Assignments

Unit 8 Allowance for Threaded Fittings

Unit 10 Allowances for Copper Fittings

Unit 12 Allowance for Plastic Fittings

I' Math: Units 8, 10, 12

Unit 15 Equal Spacing

Unit 16 Angles in Plumbing

I' Math: Units 15-16

Unit 18 Solving with 45° Constants

Unit 19 45° Diagonal and Pipe Length

,f' Math: Units 18-19

Unit 20 Three-Pipe Diagrams with a 45° Offset

,f' Math: Unit 20

Conversion Review

Area, Volume Review

Math Review
Plumbing 1 Math Review Quiz
Study Resources
Data Table 5-7.pdf
Hybrid Assignments
MP Chapter 15, 16, 17
Read MOPL Chapter 18: Water Heaters
MP Chapter 18 Lab Workbook
MP Chapter 18 Know and Understand
Read IPC Chapter 5 - Water Heaters
IPC SS 6 Quiz - Water Heaters
DISCUSSION: Water Heaters
Hybrid Assignments
Read Chapter 7: IPC Code Book Sanitary Drainage
IPC Chapter 7 SS 9 Quiz - Sanitary Drainage I (sections
701 - 707)
IPC Chapter 7 SS 10 Quiz - Sanitary Drainage II { sections
708 - 718)
Read Chapter 8: IPC Code Book Indirect/Special Waste
IPC Chapter 8 Worksheet - Indirect/Special Waste
Hybrid Assignments
Read IPC Code Chapter 10 Traps, Interceptors and
Separators
IPC Chapter 10 • Traps, Interceptor and Separators
Read IPC Chapter 11 & 12 Storm Drainage & Special
piping and Storage Systems
Read IPC Chapters 13-14, Non potable Water Systems &
Subsurface Graywater Soil Absorption Systems

IPC Ch 11-14 Quiz - Storm Drainage, Special Piping and
Storage Systems, Non potable Water Systems and
Subsurface landscape Irrigation Systems
Hybrid Assignments - Make up Days
Hybrid Assignments
Unit 11 Plastic Pipe
Math Unit 11 Quiz
Unit 13 Welded Steel Pipe
Math Unit 13 Quiz
Unit 14 Allowing for Welded Fittings
Math Unit 14 Quiz
Unit 17 Offset, Diagonal, Rise, and Run
Math Unit 17 Quiz
Hybrid Assignments
Unit 22 45° Offsets with a Wye Fitting
Math Unit 22 Quiz
Unit 23 Wye and Tee--Wye Assemblies
Math Unit 23 Quiz
Unit 24 The 0. 707 Constant in 45° Pipe Assemblies
Math Unit 24 Quiz
Hybrid Assignments
Final Competency Exam Review
On Campus Pass-off
Final Competency Exam
On Campus Pass-off
Competency test retake

Subject to change. Please consult your Canvas course for the most current instructions and updates.

Classroom Hours

Start: 01-05-2027

End: 04-21-2027

Tu, W

6:00 PM - 9:00 PM

For a full list of course hours visit: [Course Schedule](#)

Instructor Contact Information

Jason Parsons — jparsons@stech.edu

Wade Esplin — wesplin@stech.edu

Office Hours: By appointment

Email is the preferred method of communication; you will receive a response within 24 hours during regular business hours.

Canvas Information

Canvas is where course content, grades, and communication will reside for this course.

- Canvas URL: stech.instructure.com
 - For Canvas passwords or any other computer-related technical support contact Student Services.
 - For Regular Hours and Weekdays call (435) 586 - 2899
 - For After Hours and Weekends call (435) 865 - 3929 (Leave a message if no response)
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Policies

Course Grading

You will also be required to complete the course, exams and homework, with 75% or higher accuracy for credit and DOPL licensure points.

Grade Scale: The following grading scale will be used to determine a letter grade.

A : 94 - 100%	A- : 90 - 93%	B+ : 87 - 89%	B : 83 - 86%
B- : 80 - 82%	C+ : 77 - 79%	C : 73 - 76%	C- : 70 - 72%
D+ : 67 - 69%	D : 63 - 66%	D- : 60 - 62%	F : 0 - 59%

Course Policies

Cell phones for many have become a distraction. When you are in class or lab we encourage you to keep your cell phones put away in a secure location. If you use ear buds we ask that you only use one so you can still hear the things going on around you. If you are using your phone for things other than school related items, instructors will ask you to put them away. Please follow the direction of your instructors. Those who have been asked to refrain from using your cell phone and fail to do so will be asked to meet with the Director of Transportation, and student services will be notified.

Additional Information

InformaCast Statement: Southwest Tech uses InformaCast to ensure the safety and well-being of our students. In times of emergency, such as weather closures and delays, this app allows us to promptly deliver notifications directly to your mobile devices. To stay informed and receive real-time updates, we encourage all students to sign up for notifications. Your safety is our priority, and staying connected ensures a swift response to any unforeseen circumstances. More information and directions for signing up are available at: <https://stech.edu/emergency-notifications/>

Internet Acceptable Use Policy: The student is expected to review and follow the Southwest Technical College Internet Safety Policy at: <https://stech.edu/students/policies/>

Student Code of Conduct Policy: The student is expected to review and follow the Southwest Technical College Student Code of Conduct Policy at: <https://stech.edu/students/policies/>

Accommodations: Students with medical, psychological, learning, or other disabilities desiring accommodations or services under ADA, must contact the Student Services Office. Student Services determines eligibility for and authorizes the provision of these accommodations and services. Students must voluntarily disclose that they have a disability, request an accommodation, and provide documentation of their disability. Students with disabilities may apply for accommodations, based on an eligible disability, through the Student Services office located at 757 W. 800 S., Cedar City, UT 84720, and by phone at (435) 586-2899. No diagnostic services are currently available through Southwest Technical College.

Safety and Building Maintenance: The College has developed and follows a variety of plans to ensure the safe and effective operation of its facilities and programs. The following plans are available online: 1) Facilities Operations and Maintenance Plan; 2) Technical Infrastructure Plan; and 3) Health and Safety Plan.

Withdrawals and Refunds: Please refer to the Southwest Technical College Refund Policy at: <https://stech.edu/students/policies/>

Any high school or adult student, who declares a technical training objective is eligible for admission at Southwest Technical College (Southwest Tech). Program-specific admissions requirements may exist and will be listed on the Southwest Tech website. A high school diploma or equivalent is not required for admission but is mandatory for students seeking Title IV Federal Financial Aid.

Non-Discriminatory Policy: Southwest Technical College affirms its commitment to promote the goals of fairness and equity in all aspects of the educational enterprise, and bases its policies on the idea of global human dignity.

Southwest Tech is committed to a policy of nondiscrimination. No otherwise qualified person may be excluded from participation in or be subjected to discrimination in any course, program or activity because of race, age, color, religion, sex, pregnancy, national origin or disability. Southwest Technical College does not discriminate on the basis of sex in the education programs or activities that it operates, as required by Title IX and 34 CFR part 106. The requirement not to discriminate in education programs or activities extends to admission and employment. Inquiries about Title IX and its regulations to STECH may be referred to the Title IX Coordinator, to the Department of Education, and/or to the Office for Civil rights.

If you believe you have experienced discrimination or harassment on our campus, please contact the Title IX Coordinator, Cory Estes: cestes@stech.edu, (435) 865-3938.

For special accommodations, please contact the ADA Coordinator, Cyndie Tracy: ctracy@stech.edu, (435) 865-3944.

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